



# RAMCO INSTITUTE OF TECHNOLOGY

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## Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

**Degree, Semester & Branch** : III Semester B.E CSE  
**Course Code & Title** : CS3352 & Foundations of Data Science  
**Name of the Faculty member** : Mrs. S. Vijaya Amala Devi, AP / CSE

### Innovative Practice Description

- **Unit / Topic:** Unit V/ Data Visualization-Histogram and Legend
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TL10
- **Activity Chosen:** Think-Pair-Share
- **Justification**
  - ❖ "Think-Pair-Share" is a cooperative learning strategy commonly used in classrooms and group settings to encourage active participation, critical thinking, and discussion. Here's how it works:
  - ❖ **1. Think:** In this first stage, individuals take a moment to think silently and independently about a specific question, concept, or topic. This allows everyone to process their thoughts and form their initial ideas.
  - ❖ **2. Pair:** After the thinking stage, participants pair up with a neighbour or colleague. They share their thoughts, ideas, or answers with their partner, engaging in a discussion about the topic at hand. This step promotes collaboration and peer-to-peer learning.
  - ❖ **3. Share:** Following the pair discussion, the facilitator or group leader invites pairs to share their insights, solutions, or key takeaways with the entire group. Participants can volunteer to share their pair's discussion or be randomly selected. This stage encourages active participation, diversity of perspectives, and the synthesis of ideas.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  - ❖ Think-Pair-Share innovative practice conducted for II year CSE students, after explanation of the concept of Data Visualization with Pandas in Histogram and Legends.
  - ❖ I initiated the process by prompting the students to reflect on the concept of Histogram. Specifically, I asked them to consider the use of attributes of histogram utilized such as title, legend, figure etc., Subsequently, the students actively engaged with this concept by applying the appropriate attribute to the given dataset, aiming to determine the histogram and legend
  - ❖ After the students had completed their individual calculations, I grouped them into pairs and prompted them to communicate and explain the steps they had taken to arrive at their results to their respective partners.

- Finally, I asked any two or three pairs to explain the concept to whole class for further discussion. The students shared their ideas and knowledge of how histogram works in this program and how to calculate the averages to all students.

- CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO8 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|-----|------|------|------|
| CO5 | 3   | 3   | 3   | 1   | 1   | 1    | 2    | 3    |

(1 – Low      2 – Moderate      3 – High)

- PO / PSO mapped:**

| Innovation Practice                  | PO1                                                                                                                | PO2                                                                                  | PO3                                                                                   | PO4                                                                                       | PO12                                                                                                                                                                                  | PSO1                                                                                       |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|                                      | 2                                                                                                                  | 2                                                                                    | 3                                                                                     | 2                                                                                         | 2                                                                                                                                                                                     | 2                                                                                          |
| <b>Justification for Correlation</b> | Apply basic Knowledge and fundamentals of visualization libraries in python to interpret data for solving problems | Identify the need of visualization libraries in python to interpret and explore data | Able to design solutions for any applications using visualization libraries in python | Able to analyze and apply visualization libraries in python and provide valid conclusions | Ability to recognize the need for visualization libraries in python in real world applications and ability to apply the visualization libraries in python for developing applications | Exhibit knowledge in visualization libraries for data science for application development. |

- **Images / Screenshot of the practice:**



Figure1: Discussion with team members regarding the problem.



Figure2: Explaining the solution on board

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Collaborating in pairs enables individuals to exchange their expertise, leading to a more comprehensive grasp of the topic through shared insights. This collaborative approach enhances understanding by leveraging the collective knowledge of both individuals.

- ❖ **Feedback of practice from students and other stakeholders:**

- This activity serves as an opportunity for students to review and reinforce their understanding of Data Visualization techniques. It allows them to revisit and recall the key concepts while focusing on areas where they may be unclear.
- The activity helped the instructor to analyze, evaluate and improve their own learning.
- Various types of programming questions are deliberated in the classroom,

providing students with a comprehensive understanding of how to approach and solve them.

❖ **Benefit of the practice:**

- The students are actively involved in the learning process, and there is a mutual exchange of knowledge among them.
- Promotes an interactive classroom environment that cultivates a friendly yet academically-focused atmosphere.
- Incorporates all students into the teaching-learning process.

❖ **Challenges faced in implementation**

- Students encountered significant difficulty with this activity and struggled to complete it within the given time frame, particularly when dealing with new real-time system concepts.
- Students exhibit hesitancy when it comes to speaking in front of the entire class.

**References:**

- ❖ <http://www.et.iitb.ac.in/TeachingResources.html>
- ❖ <https://www.readingrockets.org/strategies/think-pair-share>
- ❖ <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>
- ❖ <https://serc.carleton.edu/introgeo/interactive/tpshare.html>
- ❖ [https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2.KVS\\_TPS\\_FDS.pdf](https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2.KVS_TPS_FDS.pdf)

**Signature of Faculty Member**

**HOD**